RetinAI ROP Staging: A Results Compendium of Every Model Trained (2026)

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***Abstract*—This document reports, without discussion, every model the RetinAI project trained for staging retinopathy of prematurity (ROP) in 2026, the protocol each was scored under, and the file each number comes from. The current generation (9 arms, A–I) shares one pre-registered protocol: patient-grouped five-fold cross-validation (CV) on 6,700 photographs of 1,528 infants from 3 hospitals, eye-level scoring at a fourth hospital unseen in training, and leave-one-source-out folds. The selected model, B (DINOv2 ViT-S/14), scores CV QWK5 0.825 [0.737, 0.911] and, at the unseen hospital, eye-level QWK5 0.756 [0.679, 0.821]. Earlier generations are tabulated separately under their own protocols, as are three independent packages a team member trained in November 2026 on a photo-level split of 3,141 of the same photographs (hand-made features with classical learners, ImageNet CNNs, and a RETFound-DINOv2 adapter model; best four-class test accuracy 0.860), with the reasons their numbers are not comparable.**

***Index Terms*—retinopathy of prematurity, ICROP-3, staging, benchmark, results, pre-registration**

I. How to Read

Settings. (i) *CV per photograph*: five patient-grouped folds, out-of-fold, seeds 42 and 1337 pooled, 14 duplicates excluded. (ii) *Unseen hospital per eye*: Multi-View development half (663 photographs, 230 eyes); eye = most severe predicted stage; descriptive; locked half unopened for A–I. (iii) *LOSO*: train on two sources, score the third. **Metrics.** QWK5: quadratic-weighted kappa, five ordinal levels, AP-ROP folded onto stage 4/5 [1]; mF1: mean per-class F1, present classes; Acc.: exact six-class matches; CIs: 95% percentile bootstrap, 2,000 infant resamples. **Pre-registration.** `PROTOCOL_V2.md` (`ac33f29`, 2026-10-20; 5 amendments to 2026-10-31); selection on CV only. **Caveats** (`MODEL_HISTORY.md`): QWK definitions, eye rules and test halves changed across generations; the June binary numbers carry a device shortcut. Sections III and IV are never pooled with A–I. Table I lists every generation's best model under its own protocol with the reason it is or is not comparable.

II. Current Generation (Protocol v2, Arms A–I)

Tables II–IX and Figs. 1–9. Arms: A RETFound-Green [2], B DINOv2 [3], C EfficientNetV2-S [4] from [5], D–I variants of B (Table VIII).

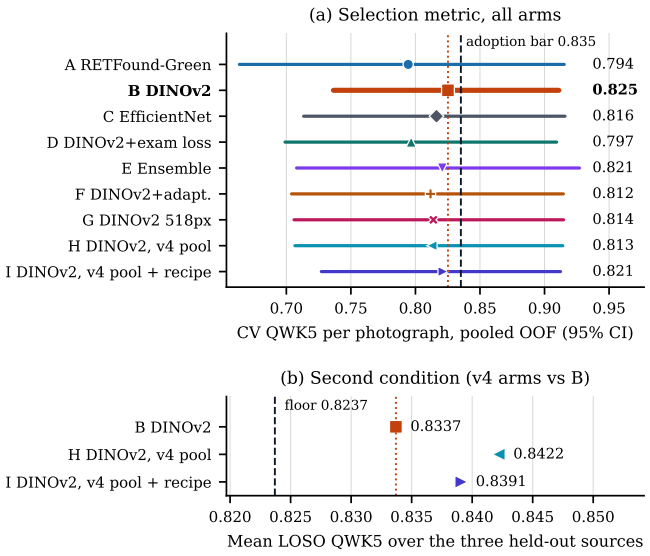


Fig. 1. Selection. (a) CV QWK5 per photograph, pooled out-of-fold, 95% CI. Dotted: B, chosen by the v3 rule. Dashed: the Amendment 5c adoption bar (B + 0.010) a v4 arm (H, I) had to reach. (b) Its second condition: mean LOSO QWK5 over three held-out sources at least B − 0.010 (dashed). No v4 arm met both; B stays.

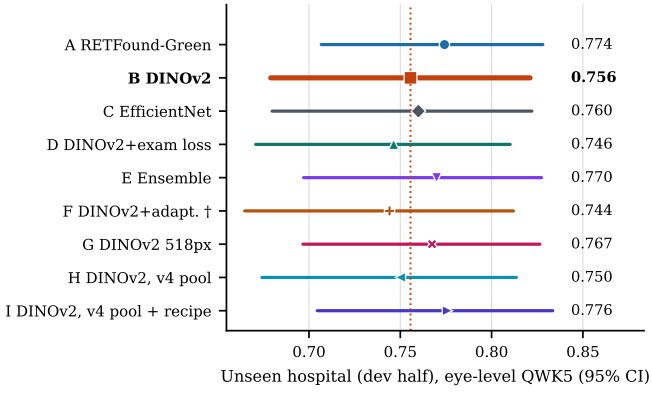


Fig. 2. Unseen hospital, development half: eye-level QWK5, 95% CI, 10 fold-seed models per arm. Descriptive; selected nothing. †Transductive.

TABLE I

EVERY GENERATION AT A GLANCE: THE BEST MODEL OF EACH UNDER ITS OWN PROTOCOL. ONLY ONE ROW IS THE PROJECT'S SELECTED MODEL; EVERY OTHER ROW IS LISTED WITH THE REASON ITS NUMBER CANNOT BE READ AGAINST IT. TEAMMATE ROWS: FOUR CLASSES, PHOTO-LEVEL SPLIT, TEST N = 315 (629 FOR ROP_ML); CI: 95% WILSON.

Generation	Best model	Task; split; unit	Headline, own protocol	Comparable with B?
v1, Jun 2026	EfficientNet-B3, one model for ROP and DR	ROP yes/no; image-level split; photo	test acc. 0.733, ROP F1 0.706	No: image-level split; other task (Table X)
Jun 2026 sweep	ResNet50	ROP yes/no; Ostrava; photo	test AUC 0.927, QWK_2 0.688	No: device shortcut, AUC 0.911 from image size alone (Table XI)
v2, Aug 2026	EfficientNetV2-S, structured head	6 classes; patient-grouped 5-fold CV; photo / eye	CV QWK_6 0.761; locked eye $QWK5_d$ 0.813 (seed 42)	No: own QWK definitions and checkpoint rule; re-scored as arm C (Table XII)
v3/v4, Oct 2026	**B: DINOv2 ViT-S/14, 5-fold ensemble (selected)**	6 classes; protocol v2; photo / eye	CV QWK5 0.825 [0.737, 0.911]; unseen-hospital eye QWK5 0.756 [0.679, 0.821]	Reference (Table II); the served model
Teammate, Nov 2026	rop_ml: SVM (RBF) on 428 hand-made features	4 classes; photo-level 80/20; photo	test acc. 0.585, QWK 0.547	No: photo-level split (73% of test photos share an infant with training); trained on 390 dev-half and 331 locked-half photos; four classes (Table XVII)
	rop_cnn: resnest50d	4 classes; photo-level 80/10/10; photo	test acc. 0.835, QWK 0.898	As above (Table XVIII)
	rop_foundation: RETFound-DINOv2 ViT-L/14 + LoRA, 3 seeds averaged	4 classes; photo-level 80/10/10; photo	test acc. 0.844, QWK 0.900	As above (Table XIX)
	Best of the packages: 3 seeds + 4 CNNs averaged	4 classes; photo-level 80/10/10; photo	test acc. 0.860 [0.820, 0.896], QWK 0.908, AUC 0.972	As above; selected on validation, 314 photos (Table XIX)

TABLE II

HEADLINE RESULTS, ALL ARMS, BOTH SEEDS POOLED. CV: POOLED OUT-OF-FOLD, PER PHOTOGRAPH. DEV: UNSEEN HOSPITAL, DEVELOPMENT HALF, PER EYE. QWK5 WITH 95% INFANT-CLUSTERED CI; ACC. SIX-CLASS ACCURACY; BAL. ACC. MEAN PER-CLASS RECALL; MF1 MACRO-F1 OVER PRESENT CLASSES; AUC AND AP: ANY ROP VS NORMAL (MEAN OVER EACH ARM'S MODELS). BOLD LETTER: SELECTED ARM. †TRANSDUCTIVE: SAW THE DEV PHOTOGRAPHS UNLABELLED IN TRAINING.

		CV, per photograph (6,686 photos)					Unseen hospital, dev half, per eye (230 eyes)				
Arm	Model	QWK5 [95% CI]	Acc.	Bal. acc.	mF1	AUC	QWK5 [95% CI]	Acc.	mF1	AUC	AP
A	RETFound-Green ViT-S	0.794 [0.664, 0.915]	0.759	0.656	0.643	0.977	0.774 [0.707, 0.828]	0.632	0.563	0.935	0.984
B	DINOv2 ViT-S	0.825 [0.737, 0.911]	0.766	0.674	0.653	0.979	0.756 [0.679, 0.821]	0.630	0.566	0.937	0.984
C	EfficientNetV2-S (baseline)	0.816 [0.713, 0.916]	0.788	0.672	0.656	0.967	0.760 [0.680, 0.822]	0.624	0.573	0.943	0.986
D	DINOv2 + exam-level loss	0.797 [0.699, 0.909]	0.803	0.671	0.661	0.942	0.746 [0.671, 0.810]	0.609	0.542	0.935	0.984
E	Ensemble A+B+C	0.821 [0.708, 0.927]	0.787	0.680	0.664	0.981	0.770 [0.697, 0.827]	0.639	0.568	0.944	0.986
F†	DINOv2 + adaptation	0.812 [0.704, 0.915]	0.761	0.663	0.644	0.980	0.744 [0.665, 0.812]	0.624	0.552	0.940	0.985
G	DINOv2 at 518 px	0.814 [0.706, 0.915]	0.764	0.667	0.647	0.984	0.767 [0.697, 0.827]	0.627	0.551	0.939	0.985
H	DINOv2 on the v4 pool	0.813 [0.707, 0.914]	0.765	0.660	0.646	0.982	0.750 [0.674, 0.814]	0.620	0.556	0.941	0.985
I	DINOv2, v4 pool + v4 recipe	0.821 [0.727, 0.912]	0.750	0.645	0.618	0.982	0.776 [0.705, 0.834]	0.633	0.561	0.944	0.986

TABLE III

RECALL / PRECISION PER CLASS, BOTH SEEDS POOLED. UPPER BLOCK: CV PER PHOTOGRAPH; LOWER BLOCK: UNSEEN HOSPITAL (DEV HALF) PER EYE. CLASS COUNTS IN EACH BLOCK'S HEADER. CLASSES [1]: S STAGE, AP AGGRESSIVE ROP.

Arm	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP
CV, per photograph; n photos: Normal 3,837, S1 574, S2 952, S3 617, S4/5 89, AP 617						
A	0.912 / 0.947	0.569 / 0.293	0.584 / 0.541	0.725 / 0.848	0.876 / 0.741	0.268 / 0.760
B	0.909 / 0.959	0.583 / 0.302	0.561 / 0.583	0.722 / 0.788	0.882 / 0.668	0.389 / 0.705
C	0.951 / 0.932	0.497 / 0.369	0.604 / 0.565	0.724 / 0.810	0.876 / 0.596	0.382 / 0.801
D	0.993 / 0.893	0.475 / 0.582	0.488 / 0.646	0.712 / 0.772	0.865 / 0.553	0.490 / 0.581
E	0.943 / 0.949	0.551 / 0.346	0.620 / 0.570	0.729 / 0.847	0.910 / 0.684	0.328 / 0.748
F†	0.911 / 0.959	0.570 / 0.291	0.562 / 0.566	0.722 / 0.767	0.888 / 0.706	0.327 / 0.746
G	0.914 / 0.962	0.607 / 0.308	0.567 / 0.552	0.720 / 0.802	0.893 / 0.720	0.302 / 0.705
H	0.924 / 0.952	0.570 / 0.326	0.557 / 0.520	0.725 / 0.807	0.876 / 0.740	0.306 / 0.682
I	0.909 / 0.960	0.490 / 0.298	0.568 / 0.497	0.741 / 0.725	0.893 / 0.677	0.269 / 0.617
Unseen hospital (dev half), per eye; n eyes: Normal 50, S1 68, S2 46, S3 59, S4/5 5, AP 2						
A	0.930 / 0.620	0.521 / 0.595	0.544 / 0.621	0.585 / 0.888	0.540 / 0.725	0.650 / 0.106
B	0.948 / 0.623	0.510 / 0.607	0.546 / 0.666	0.566 / 0.867	0.580 / 0.760	0.700 / 0.088
C	0.972 / 0.633	0.404 / 0.621	0.591 / 0.544	0.598 / 0.883	0.660 / 0.704	0.850 / 0.135
D	0.956 / 0.606	0.482 / 0.591	0.517 / 0.632	0.537 / 0.867	0.540 / 0.768	0.600 / 0.067
E	0.974 / 0.627	0.491 / 0.623	0.570 / 0.625	0.586 / 0.887	0.560 / 0.718	0.650 / 0.096
F†	0.974 / 0.626	0.484 / 0.633	0.546 / 0.654	0.559 / 0.867	0.540 / 0.717	0.600 / 0.064
G	0.962 / 0.624	0.507 / 0.597	0.522 / 0.639	0.571 / 0.899	0.540 / 0.700	0.550 / 0.070
H	0.950 / 0.625	0.497 / 0.600	0.539 / 0.622	0.551 / 0.864	0.580 / 0.802	0.550 / 0.071
I	0.936 / 0.649	0.506 / 0.648	0.596 / 0.628	0.559 / 0.821	0.560 / 0.783	0.550 / 0.067

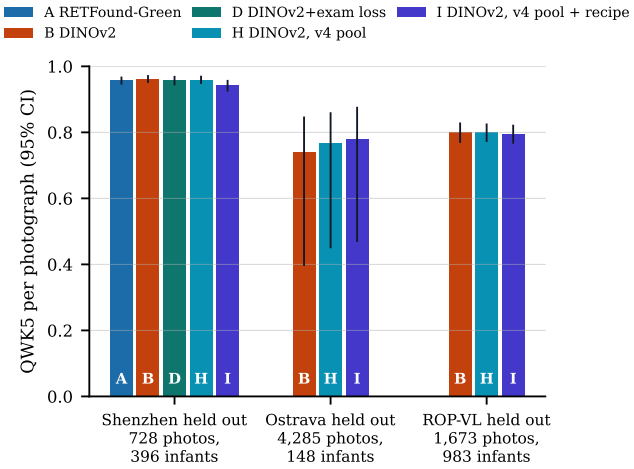


Fig. 3. LOSO QWK5 per photograph with 95% CI (Table V); bars labelled by arm. Fold compositions in Table XXII.

TABLE IV
CV QWK5 PER PHOTOGRAPH PER FOLD (F0–F4) AND SEED, EACH FOLD SCORED BY ITS OWN MODEL; MEAN AND SD OVER THE FIVE FOLDS.

Arm	Seed	F0	F1	F2	F3	F4	Mean	SD
A	42	0.623	0.868	0.872	0.907	0.918	0.838	0.122
	1337	0.595	0.868	0.882	0.911	0.916	0.834	0.135
B	42	0.743	0.870	0.853	0.904	0.898	0.854	0.065
	1337	0.666	0.879	0.857	0.923	0.909	0.847	0.104
C	42	0.680	0.865	0.889	0.899	0.926	0.852	0.099
	1337	0.667	0.863	0.853	0.934	0.917	0.847	0.106
D	42	0.691	0.851	0.810	0.897	0.882	0.826	0.083
	1337	0.650	0.858	0.816	0.920	0.883	0.826	0.105
E	42	0.682	0.874	0.905	0.918	0.929	0.862	0.102
	1337	0.634	0.881	0.878	0.933	0.935	0.852	0.125
F†	42	0.673	0.886	0.841	0.928	0.878	0.841	0.099
	1337	0.648	0.887	0.879	0.913	0.910	0.847	0.112
G	42	0.666	0.876	0.876	0.919	0.895	0.846	0.103
	1337	0.646	0.882	0.875	0.916	0.915	0.847	0.114
H	42	0.663	0.880	0.832	0.915	0.915	0.841	0.105
	1337	0.672	0.883	0.842	0.913	0.912	0.844	0.101
I	42	0.677	0.885	0.873	0.912	0.910	0.851	0.099
	1337	0.704	0.877	0.824	0.915	0.888	0.841	0.084

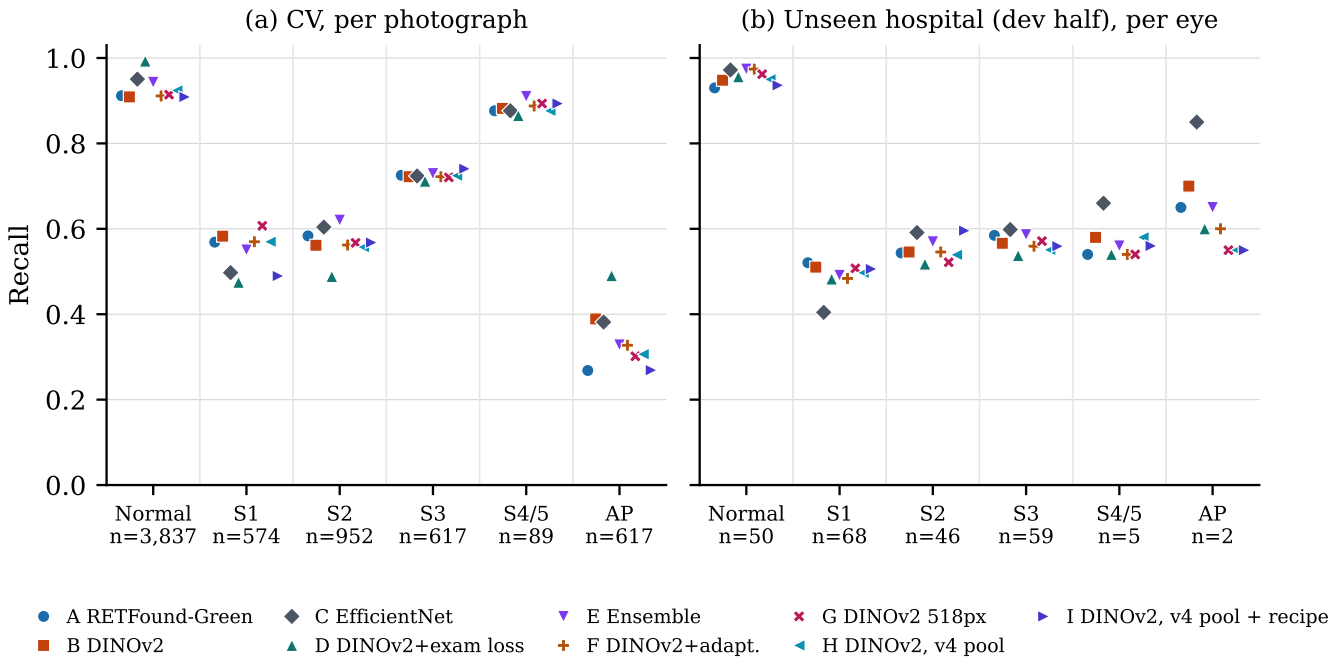


Fig. 4. Recall per class, all arms: (a) CV per photograph; (b) unseen hospital per eye. n below each class: photographs (a), eyes (b). Arm identity by colour and marker.

TABLE V

LEAVE-ONE-SOURCE-OUT (LOSO), PER PHOTOGRAPH, SEEDS POOLED: QWK5, MF1 AND ACCURACY WITH 95% INFANT-CLUSTERED CI, AND MEAN RECALL PER CLASS (– ABSENT FROM THE HELD-OUT SOURCE). SHENZHEN IS THE PROTOCOL'S SITE FOLD; THE OSTRAVA AND ROP-VL FOLDS WERE ADDED BY AMENDMENT 5 FOR B, H AND I. POOL: THE TRAINING POOL (TABLE XXI).

					Mean recall per class					
Arm	Pool	QWK5 [95% CI]	mF1 [95% CI]	Acc. [95% CI]	Normal	S1	S2	S3	S4/5	AP
Shenzhen held out (fold_site.csv): 728 photos, 396 infants, 12 duplicates excluded										
A	v3	0.958 [0.948, 0.966]	0.832 [0.799, 0.863]	0.867 [0.840, 0.892]	0.99	0.58	0.79	0.90	–	–
B	v3	0.962 [0.953, 0.971]	0.841 [0.809, 0.871]	0.882 [0.859, 0.906]	1.00	0.58	0.81	0.93	–	–
D	v3	0.958 [0.945, 0.968]	0.836 [0.804, 0.867]	0.877 [0.853, 0.900]	1.00	0.53	0.85	0.90	–	–
H	v4	0.960 [0.950, 0.969]	0.839 [0.808, 0.870]	0.877 [0.854, 0.901]	1.00	0.59	0.79	0.92	–	–
I	v4	0.942 [0.926, 0.956]	0.777 [0.741, 0.814]	0.840 [0.813, 0.868]	1.00	0.39	0.72	0.92	–	–
Ostrava held out (fold_loso_ostrava.csv): 4,285 photos, 148 infants, 0 duplicates excluded										
B	v3	0.740 [0.398, 0.845]	0.483 [0.384, 0.596]	0.791 [0.710, 0.868]	0.95	0.14	0.20	0.36	–	0.79
H	v4	0.766 [0.452, 0.858]	0.479 [0.380, 0.590]	0.797 [0.718, 0.876]	0.97	0.13	0.20	0.35	–	0.77
I	v4	0.779 [0.471, 0.875]	0.487 [0.394, 0.595]	0.804 [0.721, 0.880]	0.97	0.15	0.19	0.36	–	0.82
ROP-VL held out (fold_loso_ropvl.csv): 1,673 photos, 983 infants, 2 duplicates excluded										
B	v3	0.800 [0.771, 0.827]	0.452 [0.437, 0.467]	0.677 [0.653, 0.701]	0.94	0.74	0.60	0.71	0.00	0.00
H	v4	0.801 [0.774, 0.824]	0.443 [0.427, 0.458]	0.658 [0.634, 0.682]	0.89	0.71	0.60	0.74	0.00	0.00
I	v4	0.796 [0.768, 0.820]	0.462 [0.445, 0.479]	0.676 [0.652, 0.699]	0.94	0.48	0.67	0.81	0.00	0.06
Mean QWK5 over the three held-out sources (Amendment 5c): B 0.834, H 0.842, I 0.839										

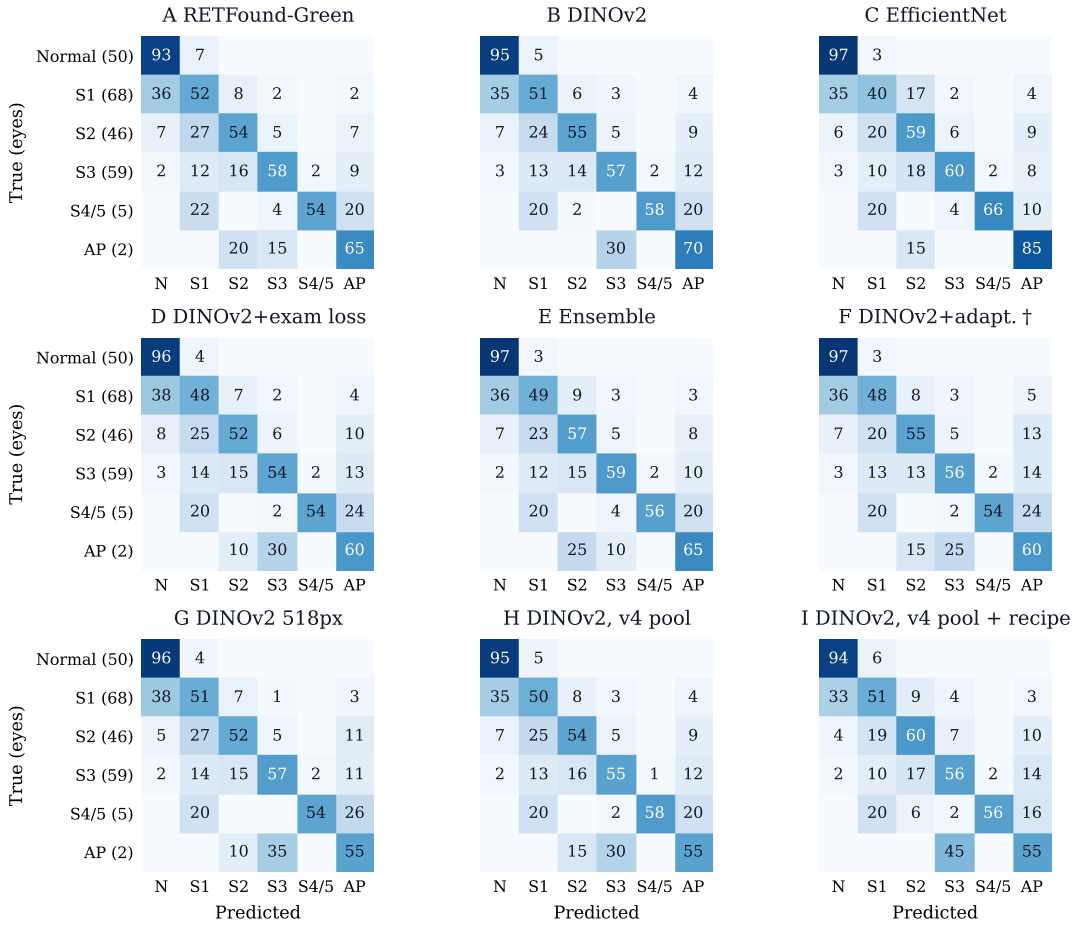


Fig. 5. Eye-level confusion matrices at the unseen hospital (dev half), all arms. Cell: share of the true-class row in %, averaged over the arm's 10 models (diagonal = recall); blank = 0. True-class eye counts in parentheses. N normal, S stage, AP AP-ROP. †Transductive.

TABLE VI

ALL 12 PRE-SPECIFIED PAIRED CONTRASTS (FIRST ARM MINUS SECOND): QWK5 AND MF1, BOTH ARMS ON THE SAME INFANT RESAMPLE, 95% CI. CV: SELECTION CONTEXT; DEV: DESCRIPTIVE. NO ROW IS A CONFIRMATORY TEST; THE PRIMARY ENDPOINT (LOCKED-HALF EYE-LEVEL A−C) IS UNSCORED.

		CV, per photograph		Dev half, per eye	
Pair	Contrast	ΔQWK5 [95% CI]	ΔmF1 [95% CI]	ΔQWK5 [95% CI]	ΔmF1 [95% CI]
A−C	retinal vs CNN (primary)	−0.022 [−0.052, +0.008]	−0.013 [−0.025, +0.012]	+0.014 [−0.012, +0.048]	−0.009 [−0.063, +0.031]
A−B	retinal vs natural	−0.031 [−0.077, +0.015]	−0.010 [−0.023, +0.036]	+0.019 [+0.000, +0.041]	−0.003 [−0.032, +0.016]
B−C	natural vs CNN	+0.009 [−0.018, +0.027]	−0.003 [−0.041, +0.018]	−0.004 [−0.038, +0.030]	−0.007 [−0.066, +0.042]
E−C	ensemble vs CNN	+0.005 [−0.007, +0.019]	+0.008 [−0.001, +0.024]	+0.010 [−0.012, +0.040]	−0.005 [−0.058, +0.034]
E−A	ensemble vs retinal	+0.026 [+0.008, +0.046]	+0.021 [+0.007, +0.029]	−0.004 [−0.020, +0.008]	+0.004 [−0.010, +0.022]
D−A	exam loss vs retinal	+0.002 [−0.057, +0.046]	+0.018 [−0.074, +0.055]	−0.028 [−0.048, −0.011]	−0.022 [−0.047, +0.019]
D−B	exam loss vs per photo	−0.029 [−0.051, +0.003]	+0.008 [−0.042, +0.038]	−0.009 [−0.019, +0.001]	−0.024 [−0.042, −0.002]
F−B	adaptation	−0.014 [−0.035, +0.008]	−0.009 [−0.021, +0.021]	−0.012 [−0.022, −0.002]	−0.014 [−0.027, +0.006]
G−B	518 vs 392 px	−0.011 [−0.036, +0.012]	−0.006 [−0.020, +0.026]	+0.012 [−0.006, +0.032]	−0.014 [−0.037, +0.016]
H−B	v4 pool	−0.013 [−0.031, +0.009]	−0.007 [−0.017, +0.019]	−0.006 [−0.015, +0.003]	−0.010 [−0.023, +0.010]
I−B	v4 pool + recipe	−0.004 [−0.015, +0.007]	−0.035 [−0.053, −0.001]	+0.020 [+0.003, +0.041]	−0.005 [−0.057, +0.033]
I−H	v4 recipe	+0.009 [−0.007, +0.020]	−0.028 [−0.043, −0.010]	+0.026 [+0.006, +0.046]	+0.004 [−0.050, +0.045]

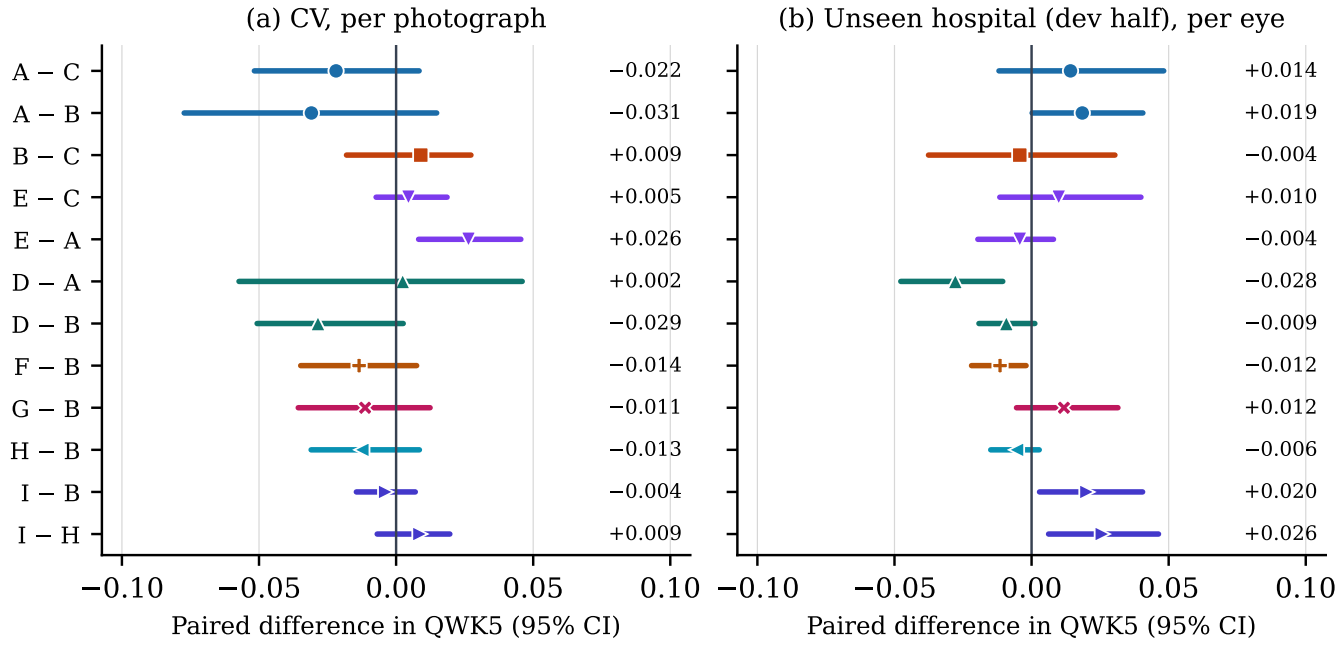


Fig. 6. Paired contrasts in QWK5 with 95% CI (Table VI): (a) CV, 6,686 photographs; (b) dev half, 230 eyes. A line crossing zero is a tie; point differences at right.

TABLE VII

OPERATING POINT AND CALIBRATION, PER ARM AND SEED. THRESHOLD: THE LARGEST VALUE OF THE UNIT ANY-ROP SCORE, $\max_i(1 - p_{i,\text{normal}})$, THAT KEEPS CV UNIT-LEVEL ANY-ROP SENSITIVITY AT OR ABOVE 0.90; CARRIED UNCHANGED TO THE DEV HALF, WHERE SENSITIVITY AND SPECIFICITY ARE MEANS OVER THE SEED'S FIVE FOLD MODELS. SEVERE: STAGE 3 OR WORSE, INCLUDING AP-ROP, AT THE SAME THRESHOLD. T: TEMPERATURE FITTED ON CV OUT-OF-FOLD UNIT VECTORS. ECE: TEN BINS, DEV EYES, SEEDS POOLED, AS TRAINED AND AFTER T (FIG. 7).

			Any ROP vs normal				Severe (S3+) vs rest					Dev-eye ECE	
Arm	Seed	Threshold	CV sens	CV spec	Dev sens	Dev spec	CV sens	CV spec	Dev sens	Dev spec	T	as trained	with T
A	42	0.9884	0.900	0.999	0.650	1.000	0.952	0.652	0.924	0.659	0.840	0.152	0.205
	1337	0.9887	0.900	0.996	0.670	1.000	0.955	0.651	0.933	0.640	0.853		
B	42	0.9894	0.900	0.996	0.656	1.000	0.945	0.648	0.927	0.654	0.850	0.157	0.203
	1337	0.9893	0.900	0.998	0.676	1.000	0.957	0.654	0.942	0.638	0.851		
C	42	0.9850	0.900	0.991	0.680	1.000	0.948	0.646	0.894	0.613	0.917	0.183	0.208
	1337	0.9805	0.900	0.993	0.671	1.000	0.942	0.645	0.918	0.633	0.935		
D	42	0.9893	0.900	0.998	0.656	1.000	0.952	0.651	0.900	0.643	0.828	0.185	0.238
	1337	0.9876	0.900	0.999	0.639	1.000	0.948	0.650	0.909	0.665	0.838		
E	42	0.9866	0.900	1.000	0.650	1.000	0.945	0.650	0.912	0.654	0.747	0.135	0.214
	1337	0.9848	0.900	0.998	0.662	1.000	0.941	0.647	0.918	0.643	0.762		
F†	42	0.9891	0.900	0.996	0.679	1.000	0.944	0.647	0.921	0.626	0.853	0.184	0.223
	1337	0.9890	0.900	0.997	0.669	1.000	0.959	0.654	0.933	0.642	0.863		
G	42	0.9882	0.900	0.996	0.643	1.000	0.949	0.649	0.921	0.665	0.854	0.174	0.210
	1337	0.9897	0.900	0.996	0.667	1.000	0.955	0.651	0.939	0.646	0.868		
H	42	0.9884	0.900	0.996	0.658	1.000	0.946	0.648	0.918	0.648	0.836	0.167	0.215
	1337	0.9898	0.900	0.998	0.661	1.000	0.953	0.652	0.927	0.648	0.851		
I	42	0.9812	0.900	0.998	0.676	1.000	0.937	0.646	0.912	0.626	0.620	0.061	0.196
	1337	0.9816	0.900	0.998	0.680	1.000	0.938	0.646	0.918	0.623	0.631		

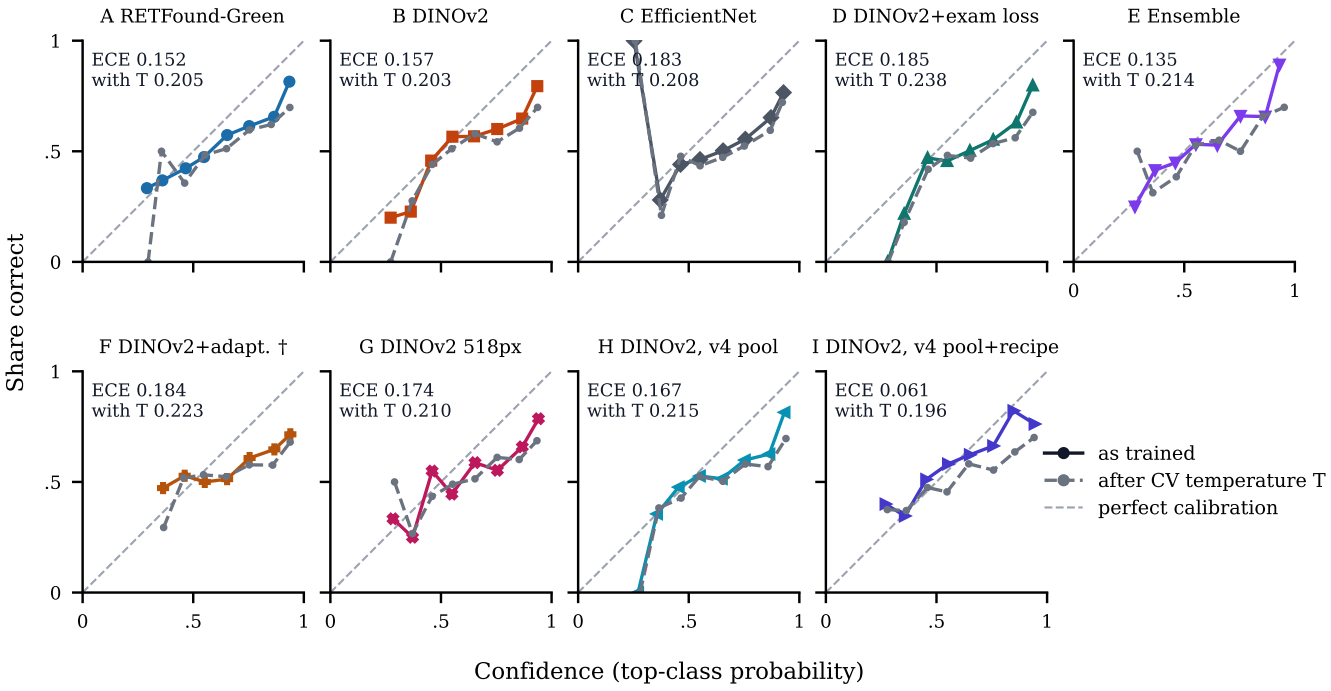


Fig. 7. Reliability of the eye-level top-class probability at the unseen hospital (dev half), ten equal-width bins, seeds pooled: as trained (colour) and after each seed's CV temperature T (grey). †Transductive.

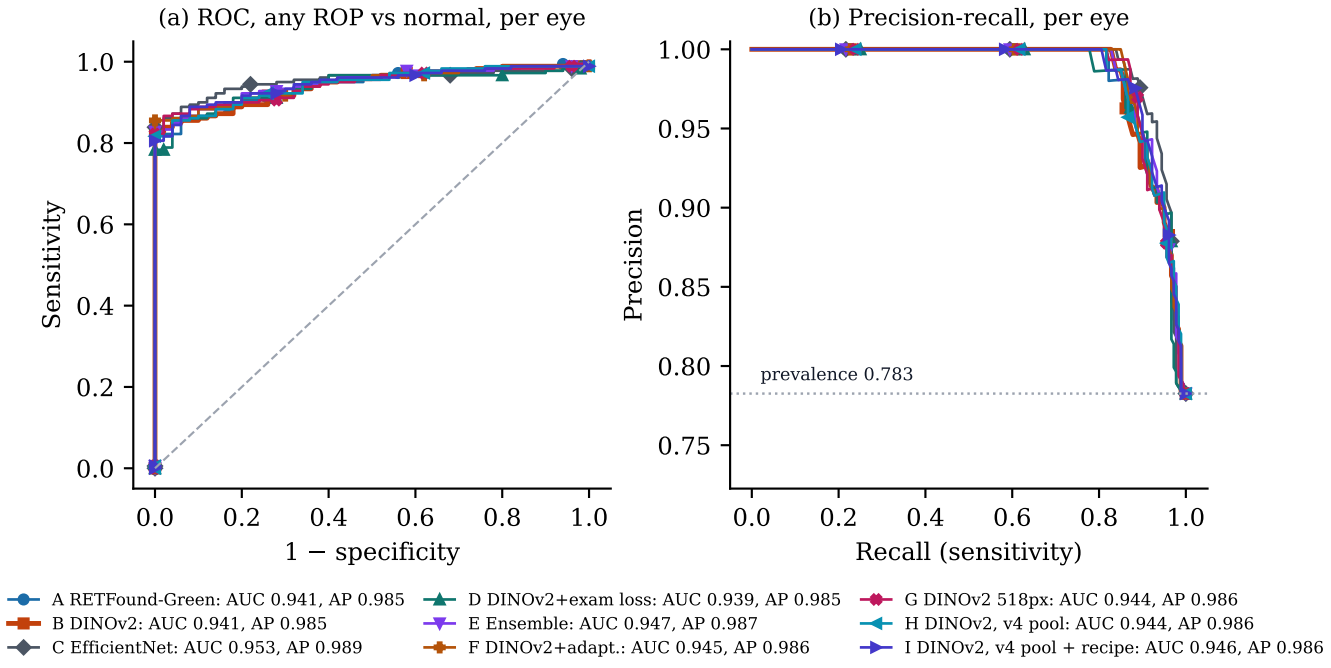


Fig. 8. Any ROP vs normal per eye at the unseen hospital (dev half), all arms. Curves use the eye score averaged over each arm's 10 models, so their AUC and AP (legend) differ slightly from Table II, which averages per-model values. Dotted: prevalence.

TABLE VIII

RECIPE AND COST. SHARED BY EVERY TRAINED ARM: CIRCLE-CROP, LETTERBOX, CLAHE; CLASS-BALANCED CROSS-ENTROPY [6]; STRONG AND SITE-RANDOMISING AUGMENTATION; BATCH 16; ADAMW; ONE-CYCLE COSINE SCHEDULE; LAST-EPOCH WEIGHTS. LLRD: LAYER-WISE LR DECAY; LS: LABEL SMOOTHING. F'S ADAPTATION IS SELF-ENSEMBLING [7]. COST: MEAN OVER CV FOLDS 0–4 AND BOTH SEEDS ON KAGGLE TESLA T4 GPUS (KERNEL LOGS); C AND E WERE NOT TRAINED IN THIS GENERATION.

Arm	Backbone	px	Norm.	Pool	Peak lr	LLRD	LS	Ep.	Differences from B's recipe	min/fold	img/s
A	`retfound_green`	392	half	v3	2e-4	0.8	0.05	20	RETFound-Green weights	32.1	38.6
B	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	reference recipe	33.3	31.9
C	`effnetv2s`	384	imagenet	v3	3e-4	–	0.05	20	v2 flat softmax, natural sampler; saved predictions re-scored	–	–
D	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	LSE bag loss on Ostrava sessions, K=8, τ=1	32.5	29.0
E	–	–	–	v3	–	–	–	–	mean of A, B, C probabilities, matched by seed; no training	–	–
F†	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	self-ensembling on unlabelled dev photos, EMA 0.99, conf. $\geq$0.9	61.7	14.2
G	`dinov2_s`	518	imagenet	v3	1e-4	0.8	0.05	20	518 px	58.1	19.2
H	`dinov2_s`	392	imagenet	v4	1e-4	0.8	0.05	20	v4 pool (per-device quality filter)	34.1	31.2
I	`dinov2_s`	392	imagenet	v4	1e-4	0.8	0.1	20	EMA weights 0.999; 1 head-only epoch; label smoothing 0.1; rot. $\pm$180°, crop $\geq$0.85, gamma, noise, Fourier swap p 0.5	34.8	32.1

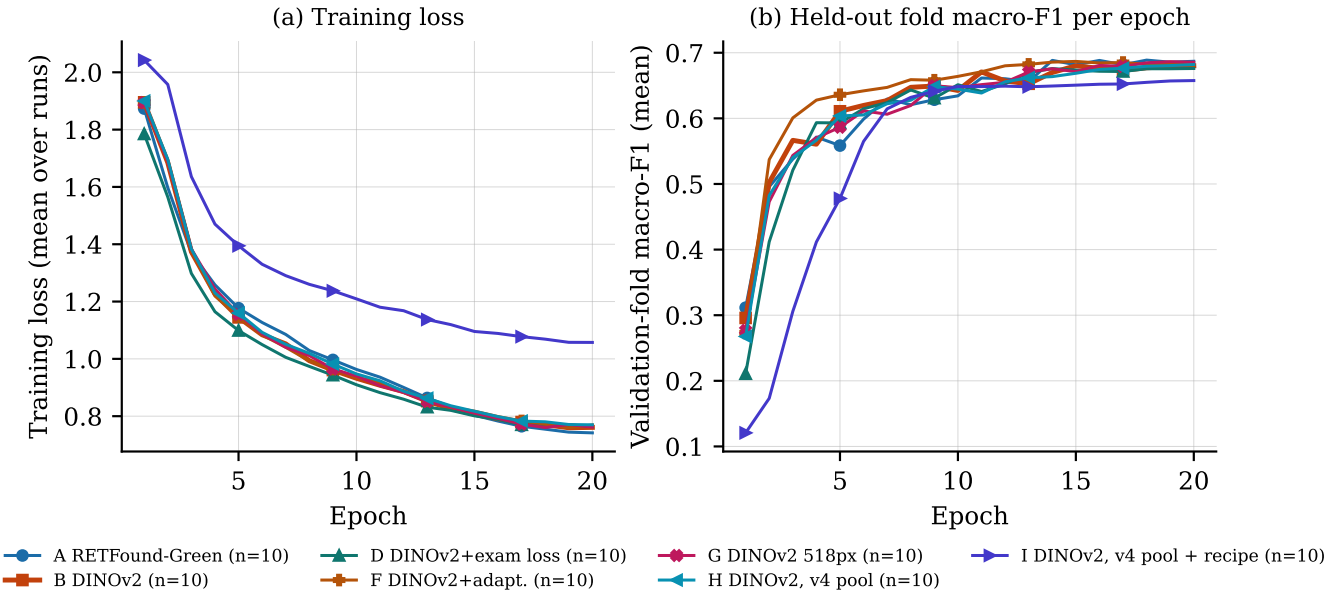


Fig. 9. Training curves, mean over each arm's 10 runs (five CV folds × two seeds): (a) training loss; (b) macro-F1 on the held-out fold per epoch. I's loss sits higher by construction (its label smoothing, Table VIII) and its F1 is measured on EMA weights. C and E have no curves.

TABLE IX

SECONDARY READOUTS. DEV PER PHOTOGRAPH: QWK5 WITH 95% CI, NO EYE AGGREGATION. CV PER COHORT AT ITS LABEL UNIT: OSTRAVA EXAMINATION SESSION, ROP-VL EYE EXAMINATION, SHENZHEN PHOTOGRAPH [5]. ORDINAL DECODER (AMENDMENT 3B): EXPECTED STAGE CUT BY FOUR THRESHOLDS FITTED ON CV ONLY; ARGMAX REMAINS PRIMARY.

	Dev half	CV, each cohort at its label unit			Ordinal decoder	
Arm	per photo QWK5 [95% CI]	Ostrava exam	ROP-VL eye-exam	Shenzhen photo	CV	Dev eye
A	0.407 [0.336, 0.478]	0.679	0.931	0.969	0.757	0.761
B	0.416 [0.342, 0.488]	0.701	0.930	0.967	0.788	0.760
C	0.438 [0.362, 0.510]	0.775	0.921	0.967	0.815	0.771
D	0.412 [0.339, 0.484]	0.846	0.925	0.967	0.808	0.762
E	0.418 [0.346, 0.492]	0.769	0.940	0.971	0.812	0.776
F†	0.416 [0.342, 0.490]	0.693	0.935	0.967	0.783	0.748
G	0.413 [0.343, 0.487]	0.695	0.925	0.967	0.779	0.764
H	0.412 [0.338, 0.486]	0.710	0.933	0.968	0.789	0.762
I	0.441 [0.363, 0.519]	0.712	0.933	0.962	0.807	0.777

TABLE X
v1 (JUNE 2026): EFFICIENTNET-B3, ONE SHARED MODEL FOR BINARY ROP AND DR. P / R: PRECISION / RECALL. *CAVEAT*: IMAGE-LEVEL SPLIT AND TWO DISEASES IN ONE MODEL; RETIRED.

Split	n	Acc.	mF1	No ROP P / R	ROP P / R	ROP F1
ROP val	928	0.647	0.646	0.476 / 0.933	0.941 / 0.510	0.661
ROP test	1,502	0.733	0.731	0.644 / 0.913	0.891 / 0.585	0.706
DR val, ep. 5	–	0.719	0.497	–	–	–

TABLE XI
JUNE 2026 BINARY SWEEP (ROP YES/NO, OSTRAVA, SWEEP VALIDATION SPLIT; QWK_2 BINARY KAPPA), AND, *, A SEPARATE RESNET50 RUN ON ITS OWN VALIDATION AND TEST SPLIT (`RESULTS/ROP/METRICS.JSON`). *CAVEAT*: INFLATED BY A DEVICE SHORTCUT (IMAGE DIMENSIONS ALONE PREDICTED ROP WITH AUC 0.911).

Architecture	Acc.	mF1	QWK_2	AUC	Params (M)	CPU ms
efficientnet_b0	0.758	0.757	0.525	0.867	4.01	56.2
efficientnet_v2_s	0.706	0.701	0.431	0.870	20.18	130.1
mobilenet_v3_large	0.744	0.743	0.500	0.857	4.2	34.4
resnet50	0.886	0.882	0.766	0.964	23.51	157.2
densenet121	0.764	0.764	0.537	0.887	6.96	136
resnet50* val	0.747	0.740	0.499	0.842	–	–
resnet50* test	0.844	0.843	0.688	0.927	–	–

III. EARLIER GENERATIONS (HISTORICAL)

Each table keeps its own protocol and definitions; none is comparable with Tables II–IX without its caveat. Glaucoma and diabetic-retinopathy models (`MODEL_HISTORY.md`) are outside this ROP compendium.

IV. INDEPENDENT TEAMMATE PACKAGES (PHOTO-LEVEL SPLIT, FOUR CLASSES)

Three packages trained by a team member on one Windows GPU (3–5 Nov 2026), outside the protocol, collected from their result files (`scripts/collect_teammate_packages.py`, `424ad77`); no model was re-run. **Data** (Table XVI): 3,141 photographs of this project (ROP-VL 1,438, Multi-View 896, Shenzhen 630, 177 renamed copies not matched by file name) in four classes (Normal, stage 1, 2, 3; no stage 4/5, no AP-ROP), split 80/10/10 *by photograph* with seed 42 (rop_ml: 80/20), and the same preprocessing in all three (black-border crop, resize, CLAHE). **Why these numbers are not comparable with Tables II–IX.** (i) The split is by photograph: 73% of the test photographs and 73% of the validation photographs share an infant with the training set (99% of the Multi-View test photographs), so every test number is optimistic by an unmeasured amount. (ii) 390 development-half and 331 locked-half photographs were used for training, so none of these models can ever be scored on this project's test halves. (iii) Four classes, 315 test photographs: a 95% CI about ± 4 points, wider than most differences between runs. (iv) Early stopping and ensemble membership were chosen on 314 validation photographs.

rop_ml (Table XVII): 428 hand-made features per 256-px photograph (colour statistics, LBP, GLCM, HOG, Frangi vessel responses), training rows balanced by augmentation, twelve learners ranked by five-fold CV accuracy; best SVM (RBF), test accuracy 0.585, QWK 0.547. **rop_cnn** (Table XVIII): six ImageNet-pretrained timm CNNs at 224 px, head warm-up then full fine-tuning, AdamW, cosine schedule, class-weighted cross-entropy with label smoothing, early stopping on validation macro-F1, three-view flip TTA; best resnest50d, test accuracy 0.835, QWK 0.898; convnext_tiny diverged (0.530). **rop_foundation** (Table XIX): RETFound-DINOv2 ViT-L/14 [8] kept frozen (309.7 M weights) with LoRA adapters [9] of rank 16 on the attention and MLP projections of all 24 blocks and a linear head (6.30 M trainable), bf16, 4-view flip TTA; a stage-2 sweep of one change each (10 epochs) chose the 280-px variant on validation, trained for 30 epochs with three seeds: test accuracy 0.787–0.832 (mean 0.814, SD 0.023), against 0.825 for the attention-only adapters (2.37 M) and 0.651 when four changes were combined. Averaging the three seeds gives 0.844 (QWK 0.900); adding the four CNNs gives the best number in the packages, 0.860 [0.820, 0.896] (QWK 0.908, AUC 0.972). Per-class recall and precision of the best models: Table XX; all 33 models with Wilson CIs: Fig. 10. **Reading.** Within the packages, a frozen retina-pretrained ViT-L with 6.30 M adapted weights ties fully fine-tuned CNNs, and seed-to-seed spread exceeds every single-change effect in the sweep; averaging models is the only reliable gain. The three-seed RETFound-DINOv2 ensemble is shown in the webapp beside B as a comparison only; B remains the selected model.

TABLE XII

v2 (August 2026): EfficientNetV2-S structured head (CORN ordinal + AP-ROP branch + site adversary, 20,512,600 parameters) vs flat softmax, natural sampling, five-fold CV. QWK_6: legacy six-class definition; $QWK5_d$: v2's "corrected" definition, rows with true or predicted AP-ROP removed (optimistic); QWK5: the v3 definition used in Table II. Dev and locked: unseen hospital; eye = max over views; locked half opened once (2026-08-18); dev and locked values are means over the five fold checkpoints. The served fold-0 checkpoint (structured, seed 42) scored per photograph $QWK5_d$ 0.493, accuracy 0.546, any-ROP AUC 0.835, and per eye $QWK5_d$ 0.839 [0.768, 0.890], accuracy 0.678, mF1 0.616. *Caveat*: arm C re-scores the flat arm under v3 rules; v2's headline eye number was on the locked half, v3's on the dev half.

		CV pooled OOF, 6,700 photos						Dev	Locked, photo		Locked, eye (max)		
Head	Seed	QWK_6	$QWK5_d$	QWK5	Acc.	mF1	AUC	QWK5	$QWK5_d$	QWK5	$QWK5_d$	QWK5	Acc.
Structured	42	0.761	0.897	0.807	0.786	0.652	0.969	0.755	0.473	0.517	0.813	0.823	0.646
	1337	0.783	0.905	0.824	0.804	0.666	0.967	0.752	0.456	0.500	0.794	0.808	0.627
Flat	42	0.777	0.897	0.820	0.789	0.655	0.968	0.756	0.492	0.551	0.808	0.823	0.635
	1337	0.770	0.898	0.813	0.787	0.658	0.966	0.764	0.486	0.543	0.801	0.818	0.630

TABLE XIII

v2 five-backbone staging sweep (flat softmax, six classes, fold 0 and the leave-site fold, seed 42). *Caveat*: one fold and one seed; the in-fold spread lies inside one run's end-of-schedule noise; legacy QWK not recomputed.

Backbone	Params	mF1, fold 0	mF1, site fold	AP-ROP recall
EfficientNetV2-S	20.2M	0.520	0.641	0.183
DeiT3-S	21.8M	0.517	0.507	0.174
ConvNeXt-T (in12k)	27.8M	0.509	0.530	0.162
CAFormer-S18	24.3M	0.498	0.507	0.221
EfficientNet-B0	4.0M	0.492	0.536	(pre-metric-fix)

TABLE XIV

Retired v2 structured arm trained with a class-balanced sampler, fold 0. *Caveat*: fold 0 only, legacy six-class QWK; the CORN cascade calibrates to the training prevalence.

Fold 0	Balanced sampler	Natural (v2)	Flat
QWK (legacy 6-class, as reported at the time)	0.121	0.569	0.594
stage_4_5 precision	0.043	—	—
Normal recall	0.456	0.939	—
macro-F1	0.403	0.553	0.563

TABLE XV

Retired ResNet50 binary screener: external audit at the unseen hospital (663 dev photographs). *Caveat*: its threshold was set at another hospital; the v2 dev AUC is in-sample for v2's own threshold choice.

External audit	Value
images flagged positive	663 of 663 — including every healthy eye
specificity at the served threshold	0.000
external AUC	0.691
minimum score among external healthy eyes	0.2939
served threshold	0.1933
external dev AUC ("any ROP") (retired / v2 structured)	0.691 / 0.821
external locked AUC ("any ROP") (retired / v2 structured)	not measured / 0.835

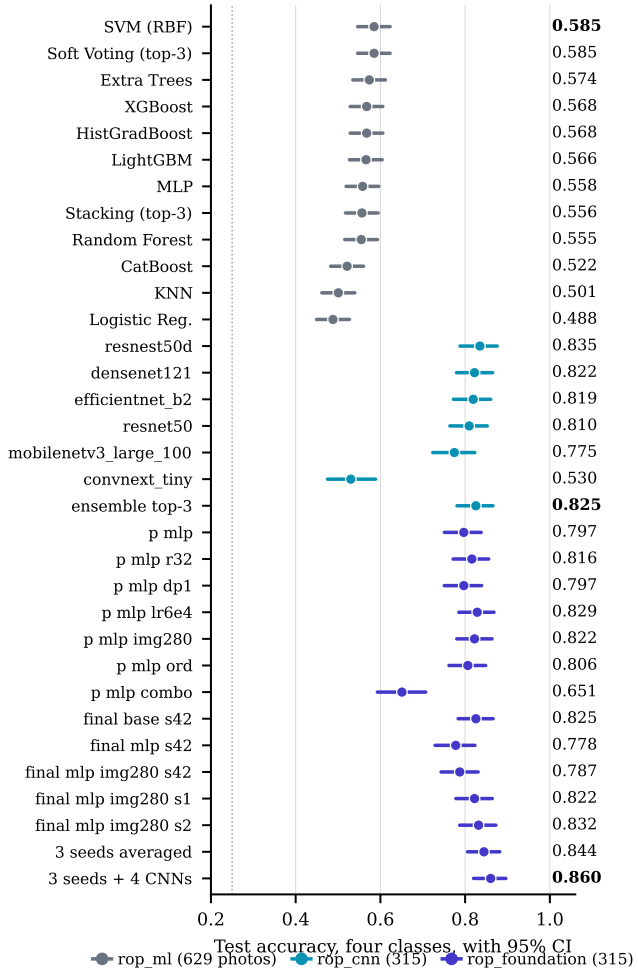


Fig. 10. Teammate packages: four-class test accuracy of every model with 95% Wilson CI (n in the legend), one colour per package. Dotted: chance. Not comparable with Figs. 1–2 (Section IV).

TABLE XVI

TEAMMATE PACKAGES: THEIR PHOTO-LEVEL SPLIT AGAINST THIS PROJECT'S SPLITS, BY FILE NAME (2,964 OF 3,141 MATCHED). "ELSEWHERE": RENAMED COPIES NOT MATCHED. LAST COLUMN: VALIDATION/TEST PHOTOGRAPHS WHOSE INFANT ALSO HAS PHOTOGRAPHS IN THEIR TRAINING SET (MATCHED PHOTOGRAPHS ONLY).

		Their classes				Where the photographs sit in this project				Same infant
Their split	Photos	N	S1	S2	S3	CV pool	Dev half	Locked half	Elsewhere	as their training
Train	2,512	687	515	651	659	1,647	390	331	144	–
Val	314	86	65	81	82	208	48	39	19	214 / 295
Test	315	86	64	82	83	213	50	38	14	221 / 301

TABLE XVII

ROP_ML: TWELVE CLASSICAL LEARNERS ON 428 FEATURES; CV: FIVE-FOLD ACCURACY ON THE (AUGMENTED) TRAINING ROWS, WHICH RANKED THE LEARNERS AND CHOSE THE TOP THREE FOR THE TWO ENSEMBLES; S: TRAINING SECONDS. *CAVEAT*: PHOTO-LEVEL SPLIT (SECTION IV).

	CV	Test, 629 photographs					
Model	Acc.	Acc.	Bal. acc.	mF1	QWK	AUC	s
SVM (RBF)	0.562	0.585	0.576	0.576	0.547	0.812	23
Soft Voting (top-3)	0.571	0.585	0.571	0.569	0.573	0.810	593
Extra Trees	0.542	0.574	0.561	0.555	0.502	0.790	6
XGBoost	0.562	0.568	0.554	0.552	0.535	0.804	130
HistGradBoost	0.566	0.568	0.552	0.550	0.551	0.807	459
LightGBM	0.567	0.566	0.555	0.555	0.563	0.810	39
MLP	0.534	0.558	0.547	0.549	0.531	0.795	13
Stacking (top-3)	0.567	0.556	0.543	0.541	0.556	0.807	2,107
Random Forest	0.540	0.555	0.540	0.532	0.500	0.786	9
CatBoost	0.547	0.522	0.511	0.508	0.458	0.784	33
KNN	0.518	0.501	0.493	0.489	0.403	0.736	5
Logistic Reg.	0.520	0.488	0.477	0.475	0.410	0.762	6

TABLE XVIII

ROP_CNN: IMAGENET-PRETRAINED CNNS (TIMM), 224 PX, UP TO 40 EPOCHS WITH EARLY STOPPING ON VALIDATION MACRO-F1; THE SOFT VOTE AVERAGES THE THREE MODELS WITH THE HIGHEST VALIDATION MACRO-F1 (EFFICIENTNET-B2, RESNET50, DENSENET121). *CAVEAT*: PHOTO-LEVEL SPLIT.

			Val	Test, 315 photographs				
Model	Par. (M)	Ep.	mF1	Acc.	Bal.	mF1	QWK	AUC
resnest50d	25.4	32	0.821	0.835	0.827	0.828	0.898	0.956
densenet121	7.0	36	0.822	0.822	0.816	0.815	0.880	0.942
efficientnet_b2	7.7	40	0.832	0.819	0.812	0.811	0.884	0.943
resnet50	23.5	40	0.827	0.810	0.802	0.800	0.860	0.951
mobilenetv3_large_100	4.2	22	0.769	0.775	0.775	0.769	0.840	0.925
convnext_tiny	27.8	13	0.491	0.530	0.546	0.520	0.497	0.821
Soft vote, top 3	–	–	0.856	0.825	0.821	0.818	0.881	0.959

TABLE XIX

ROP_FOUNDATION: RETFOUND-DINOV2 VIT-L/14 FROZEN, LORA ADAPTERS (RANK 16, ALPHA 32 UNLESS STATED) ON ATTENTION AND MLP LAYERS, LINEAR 4-CLASS HEAD; LR 4×10^{-4} UNLESS STATED; BATCH 4×4; CLASS-WEIGHTED CROSS-ENTROPY, LABEL SMOOTHING 0.1; BEST EPOCH BY VALIDATION MACRO-F1; 4-VIEW FLIP TTA. PROBES: 10 EPOCHS, ONE CHANGE EACH. *CAVEAT*: PHOTO-LEVEL SPLIT; THE CHOSEN CONFIGURATION WAS PICKED ON VALIDATION AND IS NOT BETTER THAN THE ATTENTION-ONLY RUN ON TEST.

		Trainable			Val	Test, 315 photographs				
Run	Change	(M)	px	Epochs	mF1	Acc.	Bal. acc.	mF1	QWK	AUC
Stage-2 probes, 10 epochs, seed 42 (one change each vs the first row)										
p mlp	LoRA on attention + MLP (probe base)	6.30	224	10	0.810	0.797	0.798	0.792	0.862	0.947
p mlp r32	rank 32, alpha 64	12.59	224	10	0.819	0.816	0.811	0.810	0.867	0.950
p mlp dp1	drop-path 0.1	6.30	224	10	0.830	0.797	0.799	0.793	0.861	0.951
p mlp lr6e4	lr 6e-4	6.30	224	10	0.829	0.829	0.825	0.823	0.897	0.958
p mlp img280	280 px input	6.30	280	10	0.834	0.822	0.821	0.818	0.875	0.959
p mlp ord	ordinal auxiliary loss 0.3	6.30	224	10	0.807	0.806	0.807	0.803	0.876	0.951
p mlp combo	r32 + dp 0.1 + lr 6e-4 + 280 px	12.59	280	10	0.623	0.651	0.662	0.617	0.761	0.900
Full-length runs, 30 epochs max										
final base s42	attention-only LoRA, 30 ep	2.37	224	30	0.817	0.825	0.818	0.818	0.900	0.956
final mlp s42	attention + MLP, 224 px, 30 ep	6.30	224	25	0.816	0.778	0.763	0.765	0.860	0.948
final mlp img280 s42	chosen config, seed 42	6.30	280	25	0.846	0.787	0.787	0.779	0.870	0.950
final mlp img280 s1	chosen config, seed 1	6.30	280	30	0.841	0.822	0.818	0.815	0.888	0.952
final mlp img280 s2	chosen config, seed 2	6.30	280	30	0.846	0.832	0.827	0.826	0.889	0.951
3 seeds averaged	chosen config, seeds 42, 1, 2	–	–	–	0.839	0.844	0.842	0.839	0.900	0.964
3 seeds + 4 CNNs	plus EffNet-B2, ResNet50, DenseNet121, ResNeSt-50d	–	–	–	0.863	0.860	0.859	0.855	0.908	0.972

TABLE XX

RECALL / PRECISION PER CLASS OF THE PACKAGES' BEST MODELS ON THEIR OWN TEST PHOTOGRAPHS. *CAVEAT*: PHOTO-LEVEL SPLIT; ROP_ML'S TEST SET DIFFERS FROM THE OTHER TWO.

Model	Normal	Stage 1	Stage 2	Stage 3
rop_ml test, 629 photographs; n: Normal 172, S1 129, S2 163, S3 165				
SVM (RBF)	0.767 / 0.706	0.450 / 0.423	0.515 / 0.486	0.503 / 0.629
rop_cnn and rop_foundation test, 315 photographs; n: Normal 86, S1 64, S2 82, S3 83				
resnest50d	0.977 / 0.966	0.719 / 0.742	0.793 / 0.730	0.819 / 0.883
CNN vote, top 3	1.000 / 0.915	0.781 / 0.676	0.720 / 0.756	0.783 / 0.942
RETFound LoRA, 3 seeds	0.965 / 0.965	0.828 / 0.726	0.707 / 0.806	0.868 / 0.857
3 seeds + 4 CNNs	0.988 / 0.966	0.859 / 0.733	0.744 / 0.803	0.843 / 0.921

V. Data

Ostrava [10] labels one examination session; ROP-VL one photograph with an eye-examination identifier; Shenzhen one photograph; Multi-View one eye.

VI. Reproducibility

Figures, tables and every number above: `.venv/bin/python scripts/paper3_figures.py`, then `tectonic -X compile paper3_results.tex`. No model is run and no file named `test_locked` is read. PKG = `FINAL_MODEL_EVIDENCE_PACKAGE`.

Compliance with Ethical Standards

This retrospective study used only publicly released, de-identified datasets under their licences (Ostrava/RIDIRP, ROP-VL, Shenzhen, Multi-View ROP, CC BY-NC); no new data were collected and no ethical approval was required. RETFound-Green's and RETFound-DINOv2's weights are used under their non-commercial licences. The system is research software and not a medical device. No funding was received; the authors declare no conflicts of interest.

Acknowledgments

We thank the releasing institutions and Kaggle for the free GPU time on which every model was trained. **[FILL: venue-specific AI disclosure, e.g.: Claude (Anthropic) was used to draft and edit code and text in all sections under the authors' direction; the authors verified every number against the committed result files and take full responsibility for the content.]**

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TABLE XXI

DATA BY HOSPITAL AND CLASS. CV POOL: EVERY PHOTOGRAPH LIES IN EXACTLY ONE VALIDATION FOLD. TRAINING-ELIGIBLE: AFTER PER-INFANT CAPS (TRAINING SIDES ONLY); THE V4 POOL ADDS A PER-DEVICE OSTRAVA QUALITY FILTER. *LOCKED-HALF LABELS WERE NOT READ; COUNTS FROM THE PROTOCOL.

Set	Unit	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP	Total	Infants
Cross-validation pool (every photo is in exactly one validation fold)									
Ostrava	photos	2,980	252	458	125	0	470	4,285	148
ROP-VL	photos	623	236	338	241	89	148	1,675	983
Shenzhen	photos	234	89	163	254	0	0	740	397
Total	photos	3,837	577	959	620	89	618	6,700	1,528
Training-eligible after per-infant caps, v3 pool (arms A–G)									
Ostrava	photos	270	190	195	40	0	20	715	
ROP-VL	photos	623	236	338	241	89	148	1,675	
Shenzhen	photos	216	89	163	254	0	0	722	
Total	photos	1,109	515	696	535	89	168	3,112	
Training-eligible, v4 pool (arms H, I)									
Ostrava	photos	270	173	191	40	0	20	694	
ROP-VL	photos	623	236	338	241	89	148	1,675	
Shenzhen	photos	216	88	158	252	0	0	714	
Total	photos	1,109	497	687	533	89	168	3,083	
Unseen hospital (Multi-View), never in any fold									
Dev half	photos	150	190	135	168	14	6	663	116
Dev half	eyes	50	68	46	59	5	2	230	116
Locked half*	photos	–	–	–	–	–	–	661	116
Locked half*	eyes	–	–	–	–	–	–	230	116

TABLE XXII

LOSO FOLD COMPOSITIONS: HELD-OUT SOURCE (TEST PHOTOGRAPHS AFTER DUPLICATE EXCLUSION, INFANTS) AND TRAINING PHOTOGRAPHS BY CLASS IN THE V3 POOL (A, B, D) AND THE V4 POOL (H, I).

					Training photographs						
Held out	Fold file	Test photos	Infants	Pool	Normal	S1	S2	S3	S4/5	AP	Total
Shenzhen	fold_site.csv	728	396	v3	893	426	533	280	89	167	2,388
				v4	893	409	529	280	89	167	2,367
Ostrava	fold_loso_ostrava.csv	4,285	148	v3	839	322	494	492	89	147	2,383
				v4	839	322	494	492	89	147	2,383
ROP-VL	fold_loso_ropvl.csv	1,673	983	v3	486	276	351	292	0	20	1,425
				v4	486	259	347	292	0	20	1,404

TABLE XXIII

SOURCES: PATH AND LAST COMMIT TOUCHING IT (`GIT LOG`); "GIT-IGNORED, ON DISK" FILES EXIST ONLY IN THE WORKING TREE (THE CSVS FALL UNDER THE REPOSITORY'S `*.csv` RULE).

Item	Path	Commit
Protocol and amendments (first commit)	`docs/paper/PROTOCOL_V2.md`	`ac33f29` 2026-10-20
Protocol, latest amendment	`docs/paper/PROTOCOL_V2.md`	`0e55915` 2026-10-31
Scores, arms A–I (selection, LOSO, contrasts)	`results/rop/v3/v3_scores.json`	`b827cce` 2026-11-01
Per-arm report data (tables, curves)	`results/rop/v3/report_data.json`	`b827cce` 2026-11-01
v4 pool summary	`results/prepared/rop_staging/v4/v4_summary.json`	`1efccd0` 2026-10-31
Scorer	`scripts/rop_v3_score.py`	`2bbc55a` 2026-10-31
Report-data builder	`scripts/rop_v3_report_data.py`	`d8b7ec1` 2026-10-31
v4 model report	`scripts/rop_v4_model_report.py`	`4e44033` 2026-11-02
This paper's figures and tables	`scripts/paper3_figures.py`	`0669798` 2026-11-02
Model chronology	`docs/paper/MODEL_HISTORY.md`	`11b8f5e` 2026-10-31
v2 QWK recalculation (3 definitions)	`PKG/20_AUDIT/verification_pass_2/06_VALIDATION/qwk_5class_vs_6class_recalculation.csv`	git-ignored, on disk
v2 CV summary	`PKG/07_TEST_RESULTS/metrics_summary.csv`	git-ignored, on disk
v2 locked, served checkpoint (eye CIs)	`PKG/08_EYE_LEVEL_ANALYSIS/eye_confidence_intervals.csv`	git-ignored, on disk
Five-backbone sweep	`PKG/19_COMPARISONS/retired_models/five_backbone_benchmark.md`	`aaffcf0` 2026-09-22
Weighted-sampler arm	`PKG/19_COMPARISONS/retired_models/weighted_sampler_v1_arm.md`	`aaffcf0` 2026-09-22
ResNet50 screener audit	`PKG/19_COMPARISONS/retired_models/resnet50_binary_screener.md`	`aaffcf0` 2026-09-22
June 2026 binary sweep	`results/sweep_rop.md`	`2bb8db3` 2026-06-28
ResNet50 val/test metrics	`results/rop/metrics.json`	`c042d7f` 2026-07-31
v1 EfficientNet-B3 evaluation	`junk/unified_model/evaluation_unified.txt`	git-ignored, on disk
Arm configs (A, B, D, F–I; C)	`configs/rop_staging_*.yaml`	`9c9ab65` 2026-10-31
Teammate packages, collected (Section \refsec:mate)	`results/comparison/teammate_packages.json`	not yet committed
Teammate-package collector	`scripts/collect_teammate_packages.py`	not yet committed